RAG Assignment: Baseline vs. Cross-Encoder Re-Ranking

Document: *The Laws of Cricket* (MCC, 2017 Code, 4th Edition, 2026)

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Dataset, setup and chunking

Setup

Component	Choice
Document	*The Laws of Cricket* (MCC, 2017 Code, 4th Edition, 2026): 90 PDF pages, 40,075 words, 42 Laws, 278 clauses
Bi-Encoder	`all-MiniLM-L6-v2` (384-d), FAISS `IndexFlatIP`, cosine similarity
Cross-Encoder	`cross-encoder/ms-marco-MiniLM-L-6-v2`
LLM	`Qwen2.5-3B-Instruct`, greedy decoding, same prompt for both pipelines
Top-K / Top-N	10 candidates retrieved / 3 chunks passed to the LLM

Chunking strategy

The Laws have a fixed hierarchy (Law → clause → sub-clause), so I split along it instead of using fixed-size windows:

- **Split:** by Law heading, then by numbered clause and sub-clause; over-long sub-clauses are cut at sentence boundaries.
- **Pack:** short consecutive pieces of the *same* Law are merged up to 900 characters (≈200 tokens), which keeps a rule next to its exceptions and fits the Bi-Encoder's 256-token window.
- **Header:** every chunk starts with `Laws of Cricket, Law` N `(Title)`, because many Laws reuse the same words (*ball*, *bowler*, *penalty runs*).
- **Drawback:** a boundary can separate a clause from its lead-in (the stem of 21.5 is in chunk 112, while 21.5.2 starts chunk 113).

Chunks	Characters (mean / median / max)	Tokens (median / max)	Above 256 tokens
265	735 / 780 / 972	183 / 263	5

Example chunk (Law 21.5, chunk 113):

```
Laws of Cricket, Law 21 (No ball)
21.5.2 the bowler's front foot must land with some part of the foot, whether grounded or raised on the same side of the imaginary
line joining the two middle stumps as the return crease described in 21.5.1, and behind the popping crease.  If the bowler's end
umpire is not satisfied that all of these three conditions have been met, they shall call and signal No ball.  ...
```

1 Architecture comparison

A **Bi-Encoder** runs the same transformer separately on the query and on each chunk, pools each into one 384-d vector, and scores relevance as their cosine similarity. Query and chunk never interact inside the network, so a whole chunk is compressed into 384 numbers before the question is known. In return, chunk vectors are computed once offline and searched with FAISS in milliseconds.

A **Cross-Encoder** feeds `[CLS] query [SEP] chunk [SEP]` through the network as one sequence, so self-attention links every query token with every chunk token in every layer. It can check specific conditions (negations, "unless" and "except" clauses, who the rule applies to) and outputs one relevance score per pair. The price: nothing can be pre-computed, so every (query, chunk) pair needs a full forward pass at query time. Hence the Bi-Encoder is fast but coarse, the Cross-Encoder precise but expensive, and it is used as a second stage on a small candidate set.

2 Qualitative and rank shift analysis

2.1 Overall retrieval quality (54 labelled queries)

Metric	Pipeline 1 (Bi-Encoder)	Pipeline 2 (+ Cross-Encoder)
Hit@1	0.519	**0.611**
Hit@3 (gold chunk reaches the LLM)	0.722	**0.741**
MRR@10	0.627	**0.686**
Recall@10 (same by construction)	0.852	0.852
Queries moved from rank 4–10 into the Top-3	6 of 54	

2.2 Rank changes of the failure queries

	Question	Gold chunk	Rank change	Ranked above gold (baseline)
Q4	Can a batter be stumped after leaving the crease to dodge a ball that could hurt them?	Law 39.2–39.3	#3 → #1	Law 39.1, Law 25.6–25.7
Q5	Can a fielder who left the field bowl as soon as they return?	Law 24.2	#2 → #1	another chunk of Law 24.2
Q2	How many runs is a hit over the rope worth?	Law 19.6–19.7	#9 → #7	Law 34.4–34.5, 25.6, 38.5–38.6 and 5 more
Q3	What happens to the one-run no-ball penalty if the batter is dismissed?	Law 21.11–21.14	#2 → #7	Law 21.15–21.17
L2	What happens if the bowler throws the ball while bowling?	Law 21.3	#27, never in Top-10	26 chunks, Top-3: Law 38.3, 21.6–21.7, 41.8

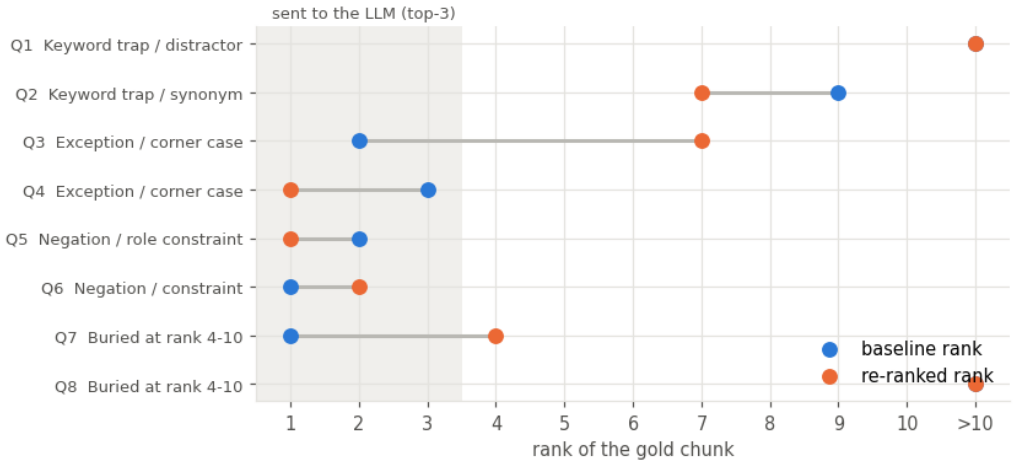


Figure 1: Rank of the gold chunk before (blue) and after (orange) re-ranking, 8 stress queries. Grey band: Top-3, sent to the LLM.

2.3 Answer 1 (baseline) vs. Answer 2 (re-ranked)

	Answer 1 (Bi-Encoder Top-3)	Answer 2 (Re-ranked Top-3)	Verdict
Q4	*No, a batter cannot be stumped after leaving their ground to avoid injury [Law 39.3.1].*	*No, a batter cannot be stumped after leaving their ground to avoid injury [Law 39.3.1]. However, if the batter leaves their ground to dodge injury and is not out stumped, they can still be out run out [Law 39.3.2].*	Both correct; A2 adds a Run out caveat.
Q5	*No, a fielder cannot bowl immediately upon returning to the field. They must serve their Penalty time, which is a maximum of 90 minutes [Law 24.2.3]. Only after serving this time can they bowl again [Law 24.2.5].*	*No, ... According to Law 24.2.5, the player shall not bowl until all of their Penalty time has been served. ... [Law 24.2.5]*	Both correct; A1 also gives the 90-minute cap.
Q2	*The provided context does not answer this question.*	*The provided context does not answer this question.*	Same abstention; gold chunk never reached the LLM.
Q3	*If a batter is dismissed, the one-run no-ball penalty shall stand even if a batter is dismissed. It shall be in addition to any other runs scored ... [Law 21.17]*	*According to Law 35.3, if the batter is dismissed while hit wicket, only the one-run penalty for a wide is awarded, and no additional 5-run penalty is given. [Law 35.3]*	A1 correct; A2 answers a different rule (wide, Hit wicket).
L2	*According to Law 38.3.1, if the non-striker is out of their ground ... the non-striker can be out Run out if the ball breaks the wicket. Therefore, if the bowler throws the ball while bowling, it could potentially put the non-striker out ... [Law 38.3.1]*	*According to Law 21.3, if either umpire considers that a further delivery by the same bowler is thrown underarm, they shall call and signal No Ball. This results in the bowler being suspended from bowling immediately ... [Law 21.3]*	A1 off-topic; A2 partly right (no warning, "underarm").

2.4 Why the baseline fails and re-ranking helps

Summary. The Cross-Encoder mainly sharpens the *top* of the list: Hit@1 rises from 28 to 33 of 54 queries and MRR@10 by 0.059. What reaches the LLM improves only slightly (Hit@3: 39 → 40; 6 queries rescued, 5 pushed out of the Top-3). On the 8 stress queries the gold clause reached the LLM in 3 cases instead of 5. Recall@10 = 0.852 caps both pipelines.

- **Q4, Q5 – re-ranking helps (modestly).** The Bi-Encoder ranked Law 39.2–39.3 third behind Law 39.1 and 25.6–25.7: all are close to "stumped" and "leaving the crease", and one cosine cannot tell that only 39.2–39.3 holds the exception "left their ground in order to avoid injury". The Cross-Encoder reads both texts together and scores it higher (2.55 vs. −0.58 and −1.00). The gold chunk was already in the Top-3, so Answer 1 omitted nothing; Answer 2 is just more complete. In Q5 only the order of two Law 24.2 chunks changed, and Answer 1 is slightly more informative.
- **Q2 – keyword trap, not fixed.** The question says "rope", the Laws say "Boundary". All Top-10 cosines are low (max 0.43) and all Cross-Encoder scores negative, so no candidate looks relevant; the gold chunk moves only #9 → #7. Both answers abstain correctly (no hallucination), but the question stays unanswered. A vocabulary gap is a first-stage problem that re-ordering the same 10 candidates cannot solve.
- **Q3 – re-ranking hurts.** The gold clause was #2 and Answer 1 stated the rule correctly. The Cross-Encoder demoted it to #7 behind chunks about wides, Hit wicket and Run out (scores 4.9–5.1 vs. 3.84) that share the words "penalty" and "dismissed". Answer 2 therefore explained the wrong rule and omitted the essential clause. Q7 (gold #1 → #4) and L1 (#3 → #5) show the same demotion; a likely cause is that the model was trained on web-search passages (MS MARCO), not on long, exception-heavy legal clauses.
- **L2 – recall ceiling.** The gold clause on thrown deliveries was only #27 of 265, so it never entered the Top-10: "bowler", "ball" and "bowling" appear in many Laws and pulled other chunks ahead. Answer 1 answered a different question (non-striker Run out). After re-ranking, a neighbouring chunk of Law 21.3 reached the Top-3, so Answer 2 mentions the bowler's suspension but omits the first-and-final warning and misstates the delivery as "underarm". Q1 and Q8 fail the same way; in Q8, Answer 1 abstained but Answer 2 invented "15 minutes" (the Law says 45).

3 Latency profiling

270 timed runs (5 × 54 queries)	mean (ms)	std (ms)	median (ms)
Pipeline 1: baseline retrieval (Bi-Encoder + FAISS)	7.9	0.5	7.9
Pipeline 2: retrieval + re-ranking	102.5	10.7	101.8
Cross-Encoder re-ranking only (10 pairs)	94.6	10.6	93.9
Slow-down of retrieval (Pipeline 2 / Pipeline 1)		13.0×	

	retrieval (ms)	LLM (ms)	end-to-end (ms)	retrieval share
Pipeline 1	7.9	691.5	699.4	1.1%
Pipeline 2	102.5	918.4	1020.9	10.0%

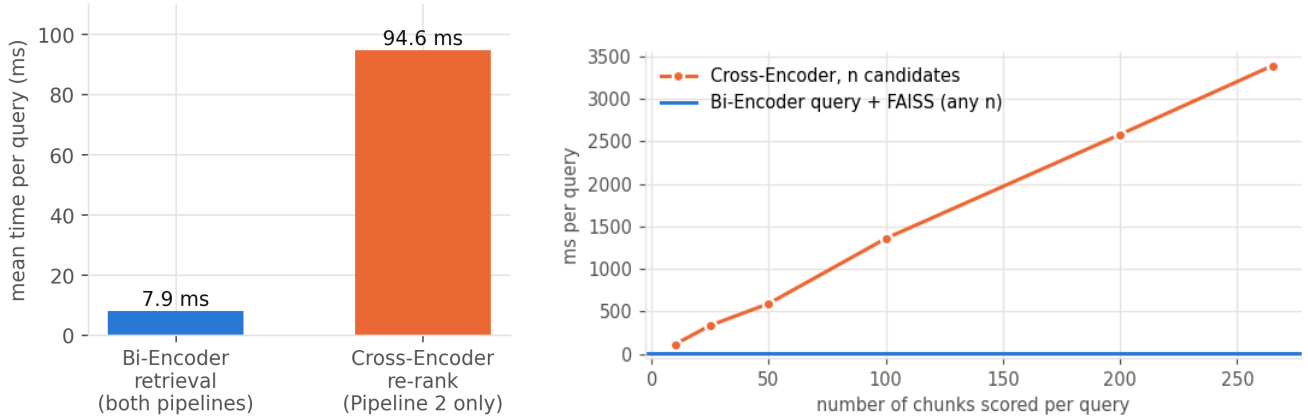


Figure 2: Left: mean time per query, Bi-Encoder retrieval (both pipelines) and Cross-Encoder re-ranking (Pipeline 2 only). Right: (a) end-to-end latency by stage; (b) Cross-Encoder cost vs. chunks scored, against the Bi-Encoder query cost.

Re-ranking raises retrieval time from 7.9 ms to 102.5 ms (13.0×); the Cross-Encoder alone accounts for 94.6 ms. Next to generation this is small: the LLM needs 0.7–0.9 s, so retrieval is 1.1% of end-to-end time in Pipeline 1 and 10.0% in Pipeline 2. The end-to-end gap (699 → 1021 ms) also includes a difference in LLM time that depends on prompt and answer length (8 queries per pipeline), so the real re-ranking cost is about 95 ms (≈14% of Pipeline 1). I find this acceptable for interactive Q&A, measured on CPU, but the return is uneven: Hit@1 +9.2 points and MRR@10 +0.059, but Hit@3 only +1.9 points. Re-ranking pays off when the top of the list matters (small Top-N), less so under a tight latency budget.

4 Scaling question

Cross-Encoder cost per (query, chunk) pair	12.8 ms
Cross-Encoder on all 265 chunks (measured)	3.4 s per query
Cross-Encoder on 10,000 / 100,000 / 1,000,000 chunks (extrapolated)	128 s / 1,280 s / 12,798 s
Exact FAISS search over 1,000,000 vectors	87 ms
Two-stage pipeline (Bi-Encoder + FAISS + Cross-Encoder on 10), any corpus size	about 136 ms
One-off indexing of 265 chunks (offline)	2.2 s
Test query: gold chunk at Bi-Encoder rank #27; Cross-Encoder over all chunks ranks it #5 in	3.3 s (two-stage: 165 ms)

The Bi-Encoder pays for document encoding once, offline; per query it needs one pass plus a vector search (87 ms exact over a million vectors). The Cross-Encoder has no reusable document representation, so each query costs one forward pass *per chunk* (12.8 ms) and grows linearly: 3.4 s for our 265 chunks, about 21 minutes for 100,000 and 3.6 hours for 1,000,000. Faster hardware changes the constant, not the growth. The two-stage design avoids this: Bi-Encoder + FAISS narrow the collection to 10 candidates and the Cross-Encoder scores only those, so its cost is fixed whatever the corpus size (only the cheap vector search grows). The trade-off is the recall ceiling: in test query L2 the gold chunk was at Bi-Encoder rank #27, outside the Top-10. Scoring all chunks would rank it #5 but takes 3.3 s instead of 165 ms (≈20× slower), and the gap widens with every added document.